ACADEMIC PROJECT REPORT

Student Progress Prediction: Machine Learning Model Analysis

A comparative study of regression models for predicting final student grades

AUTHOR Arun Sharma

PROJECT TYPE Python ML Implementation using Scikit-Learn

DATE July 2026

Project overview and research direction

Prediction objective

This project predicts students' final grades, represented by G3, from a broad set of academic, demographic, family, social, and behavioral features. The analysis compares model accuracy across Mathematics and Portuguese language courses to understand which learning patterns generalize and which are subject specific.

2
Datasets

3
Algorithms

37
Core features

G3
Target grade

Datasets

student-mat.csv covers Mathematics courses. student-por.csv covers Portuguese language courses. Both include school history, household context, study behavior, and period grades.

Algorithms and finding

Linear Regression, Decision Tree Regressor, and Random Forest Regressor were evaluated. Random Forest led on Mathematics, while Linear Regression held a slight edge on Portuguese results.

Technical stack and evaluation design

Programming language

Python 3.13.14 provides the implementation environment for data preparation, training, and evaluation.

Libraries

Pandas for data manipulation, Matplotlib for visualization, and Scikit-Learn for model training.

Development environment

VS Code supports reproducible notebooks, scripts, inspection, and iterative experimentation.

Algorithm models

LinearRegression, DecisionTreeRegressor, and RandomForestRegressor.

Evaluation framework

RMSE Root Mean Square Error. Lower values indicate smaller prediction errors.

R^2 Score Coefficient of determination. Higher values indicate stronger explanatory power.

Data source

UCI Student Performance Datasets: student-mat.csv and student-por.csv. The workflow combines categorical encoding, regression training, and held-out performance comparison.

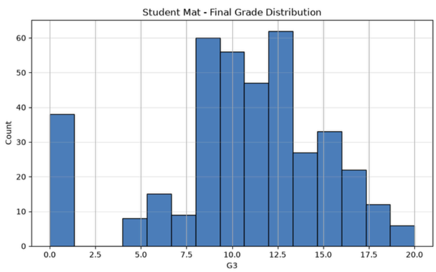
Student MAT: Mathematics profile

Dataset scope

The Mathematics dataset contains 389 students and 37 features. It combines demographic information, family relationships, parental education and employment, social behavior, health, absences, study patterns, and the period grades G1, G2, and G3.

389	37	0-20	G3
Students	Features	Grade scale	Prediction target

G3 distribution



Predictive signals

G1 and G2 are the strongest direct indicators of G3. Failures and absences provide risk signals, while study time, parental education, support services, and school engagement add contextual value.

Feature importance should be interpreted as association, not causation.

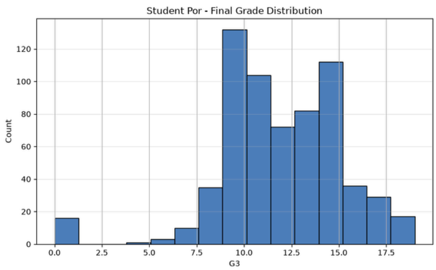
Student POR: Portuguese profile

A related but distinct learning context

The Portuguese dataset follows the same schema and records students enrolled in Portuguese language courses. It provides a larger sample than the Mathematics dataset and preserves the same broad feature families, enabling a structured comparison of subject-level model behavior.

Dimension	MAT	POR
Course	Mathematics	Portuguese
Sample size	389 students	649 students
Shared structure	37 features	37 features

POR G3 distribution



Interpretive difference

Portuguese results show a larger sample and generally lower prediction error. The broader sample may stabilize linear relationships, while subject-specific grading patterns can change the value of nonlinear interaction effects.

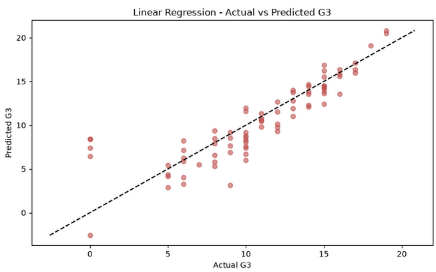
Student MAT: results and model comparison

Implementation approach

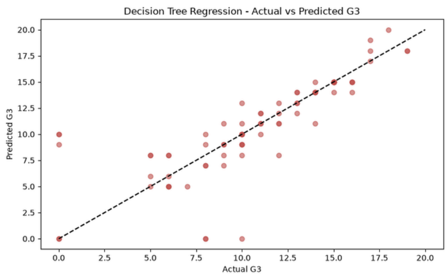
The Mathematics records were prepared for supervised regression with G3 as the response variable. Three estimators were trained and compared using RMSE and R^2 on held-out data.

Model	RMSE	R^2 score
Linear Regression	~2.38	~0.71
Decision Tree	~2.85	~0.55
Random Forest	**~1.92**	**~0.82**

Linear predictions



Decision tree predictions



Best performer: Random Forest

The ensemble reduces error to approximately 1.92 grade points and explains about 82 percent of observed variance. Its advantage suggests that Mathematics outcomes contain nonlinear interactions that a single tree or linear model does not capture as effectively.

Student POR: results and interpretation

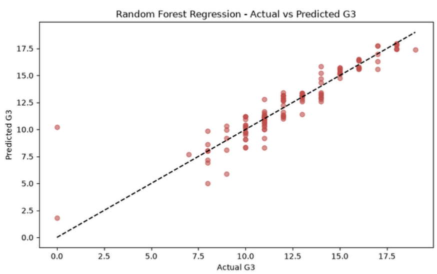
1.21
Best RMSE

0.85
Best R^2

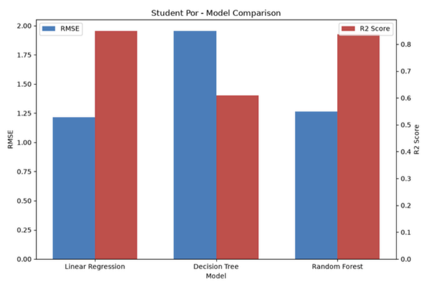
POR
Top subject model

Model	RMSE	R^2 score
Linear Regression	~1.21	~0.85
Decision Tree	~1.95	~0.61
Random Forest	~1.27	~0.84

Random Forest predictions



Model score comparison



Interpretation

Linear Regression achieves the strongest Portuguese result with an RMSE near 1.21 and R^2 near 0.85. The small edge over Random Forest indicates that the dominant relationships in this subject are comparatively stable and well represented by additive effects. Decision Tree performance is weaker and less consistent.

What the comparison tells us

0.82

Maths R^2 with Random Forest

The ensemble captured nonlinear interactions most effectively in the Mathematics dataset.

0.85

Portuguese R^2 with Linear Regression

A simpler model performed best where relationships were more stable and additive.

Conclusion

The analysis demonstrates that model choice should follow the structure of the learning context. Random Forest was the strongest option for Mathematics, while Linear Regression slightly outperformed the alternatives for Portuguese. Across both subjects, prior period grades, failures, absences, study patterns, and student context helped explain final performance.

Practical implications

Educational institutions can use calibrated predictions to identify students who may benefit from timely support, tutoring, attendance outreach, or study planning.

Future improvements

Next steps include cross validation, hyperparameter tuning, explainability analysis, fairness checks, and longitudinal data to improve reliability and responsible deployment.

Forward-looking statement

With stronger validation and careful human oversight, machine learning can become a useful decision-support layer for earlier, more individualized student success interventions.